

MIAE 221: Materials Science for Engineers

Part 7: Dislocations & Strengthening Mechanisms Master Guide

Executive Overview & Core Engineering Principles

Why is pure copper so soft that it can be bent easily by hand, yet brass (copper alloyed with zinc) or cold-worked copper can form rigid structural pipes and resilient mechanical springs? Why does blacksmithing, hammer forging, or rolling sheet metal increase its strength?

In crystalline solids:

Plastic deformation occurs through the motion (slip) of microscopic linear defects known as dislocations.

Therefore, the fundamental strategy to strengthen any crystalline metal is deceptively simple in concept, yet profound in physical application:

To make a metal stronger, you must impede, hinder, and pin the movement of dislocations!

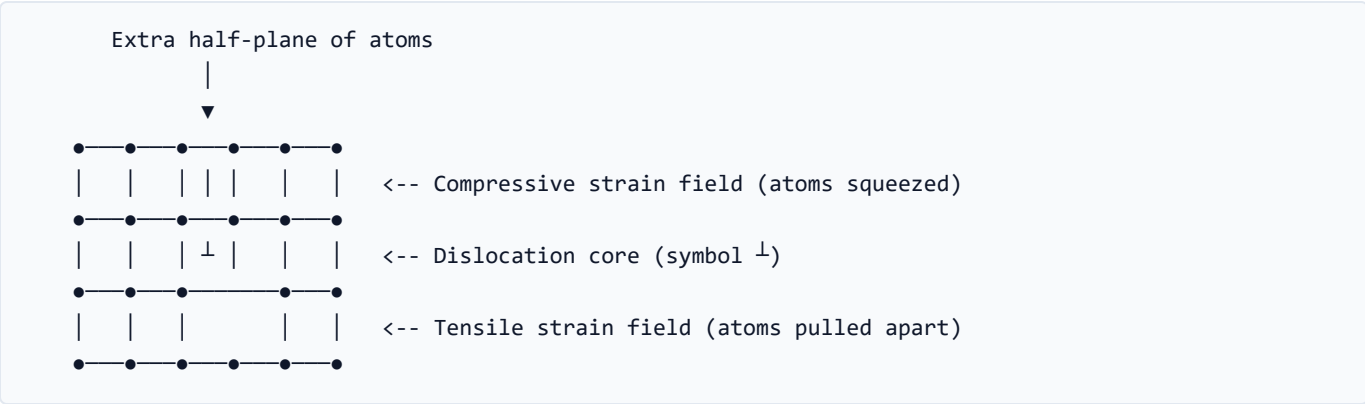
This comprehensive master guide covers:

1. **Dislocation Mechanics:** Edge, screw, and mixed dislocations; Burgers vector (**b**) orientation; dislocation glide planes; and localized elastic lattice strain fields.
2. **Slip Systems:** Close-packed crystallographic planes and directions. Complete architectural comparison of **FCC** ($111\langle 110 \rangle$, 12 systems), **BCC** ($110\langle 111 \rangle$, 48 systems), and **HCP** ($0001\langle 11\bar{2}0 \rangle$, 3 systems), explaining ductility vs. brittleness and the Taylor / Von Mises criterion.
3. **Slip in Single Crystals & Schmid's Law:** Resolved shear stress ($\tau_R = \sigma \cos \phi \cos \lambda$), Critical Resolved Shear Stress (τ_{CRSS}), and the maximum Schmid factor ($m = 0.5$).
4. **The Four Foundational Strengthening Mechanisms:**
 - **Grain Size Reduction:** Hall-Petch equation ($\sigma_y = \sigma_0 + k_y d^{-1/2}$) and grain boundary dislocation pile-ups.
 - **Solid Solution Strengthening:** Solute atom size misfit (Δr), lattice strain fields, and dislocation pinning.
 - **Strain Hardening (Cold Work):** Dislocation multiplication ($\rho_d : 10^6 \rightarrow 10^{10} \text{ cm}^{-2}$), tangling, Frank-Read sources, and percent cold work (%).
 - **Precipitation Hardening:** Orowan bowing vs. particle cutting.

- 5. **Annealing of Deformed Metals:** The three fundamental stages of thermal restoration: **Recovery**, **Recrystallization** ($T_R \approx 0.3 - 0.4, T_m$), and **Grain Growth**.
- 6. **Fully Solved Quantitative Archetypes:** Step-by-step mathematical calculations.

1. Dislocation Mechanics & Lattice Strain Fields

A dislocation is a linear (one-dimensional) crystalline defect around which lattice atoms are misaligned. Plastic deformation in metals occurs through the progressive movement of dislocations along slip planes, requiring significantly lower shear stresses (typically 10^{-4} to $10^{-3}, G$) than theoretical rigid body cleavage across an entire atomic plane simultaneously.



The Three Dislocation Archetypes

Dislocation Type	Geometry & Atomic Arrangement	Burgers Vector (b) vs. Dislocation Line (t)	Motion Direction under Applied Shear Stress
Edge Dislocation ($\perp$ or $\top$)	Extra half-plane of atoms inserted into the lattice. Atoms above the slip plane are compressed; atoms below are stretched.	Perpendicular: b $\perp$ t	Parallel to the applied shear stress vector τ .
Screw Dislocation ($\odot$ or $\otimes$)	Formed by cutting partway through the crystal and shifting one half by one lattice spacing relative to the other (spiral ramp / helical path).	Parallel: b $\parallel$ t	Perpendicular to the applied shear stress vector τ .
Mixed Dislocation	Most real dislocations exhibit curved line geometry, transitioning continuously between pure edge character and pure screw character.	At an angle θ ($0^\circ < \theta < 90^\circ$): b $\cdot$ t = $b \cos \theta$.	Net macroscopic slip is still in the direction of the Burgers vector b .

Elastic Strain Fields Around Dislocation Cores

Because atoms near dislocation cores are displaced from their equilibrium positions, localized elastic stress fields surround them:

1. Edge Dislocation:

- Region **above** the slip plane (where the extra half-plane resides): Atoms are squeezed together $\implies$ **Hydrostatic Compression** ($\sigma_{xx} < 0, \sigma_{yy} < 0$).
- Region **below** the slip plane: Atoms are pulled apart $\implies$ **Hydrostatic Tension** ($\sigma_{xx} > 0, \sigma_{yy} > 0$).

2. Screw Dislocation:

- Pure shear strain fields with zero hydrostatic volume change.

These localized strain fields interact elastically with solute impurity atoms and neighboring dislocations, forming the thermodynamic foundation for solid-solution and work-hardening mechanisms.

2. Slip Systems in Crystalline Metals

Dislocations do not glide randomly through a crystal lattice; they move preferentially along specific crystallographic planes and in specific directions that minimize the energy barrier to glide.

A **Slip System** is defined as the combination of a **Slip Plane** and a **Slip Direction**:

- **Slip Plane:** The plane of **highest planar atomic density** (closest-packed plane), which has the widest interplanar spacing (d_{hkl}), minimizing the Peierls-Nabarro friction stress.
- **Slip Direction:** The direction on that plane having the **highest linear atomic density** (closest-packed direction), corresponding to the shortest Burgers vector magnitude ($|\mathbf{b}|$).

Architectural Comparison: FCC vs. BCC vs. HCP

Crystal Structure	Slip Plane Family	Slip Direction Family	Number of Slip Systems	Ductility & Mechanical Characteristics
FCC (Cu, Al, Ni, Au, Ag, γ -Fe)	111 (4 distinct planes)	$\langle 110 \rangle$ (3 directions per plane)	12 ($4 \times 3 = 12$)	Exceptional Ductility at All Temperatures: 12 close-packed systems satisfy the Von Mises criterion (> 5). No ductile-to-brittle transition; stays tough at cryogenic temps!
BCC (W, Mo, V, Cr, α -Fe)	110 (12) 112 (12) 123 (24)	$\langle 111 \rangle$ (2 directions per plane)	48 ($12 + 12 + 24 = 48$)	High Strength, Temperature-Sensitive: No true close-packed planes; high Peierls barrier at cold temperatures leads to a dramatic Ductile-to-Brittle Transition Temperature (DBTT) .
HCP (Zn, Mg, Cd, Ti)	0001 (Basal) (1 plane)	$\langle 11\bar{2}0 \rangle$ (3 directions per plane)	3 ($1 \times 3 = 3$)	Low Ductility (Relatively Brittle): At room temperature, only 3 basal slip systems operate (fewer than the required 5 independent systems). Prone to twinning and cleavage.

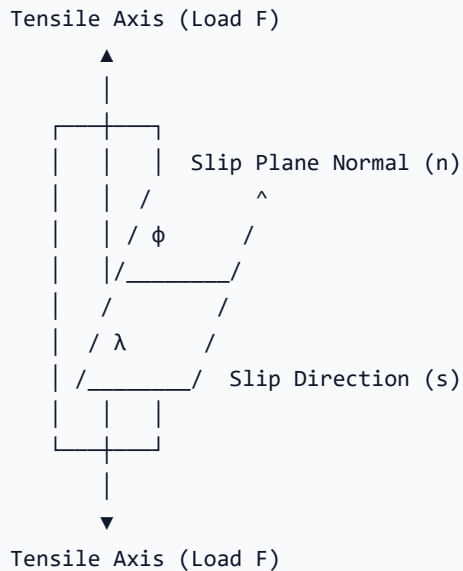
[!IMPORTANT] **Von Mises Criterion:** For a polycrystalline aggregate to undergo uniform, arbitrary plastic deformation without grain boundary cracking, each grain must possess at least **5 independent slip systems**.

- FCC easily satisfies this (≥ 5), explaining why copper and aluminum can be hammered, rolled, and drawn into extremely thin foils and wires.

- HCP has only 3 active slip systems at room temperature (< 5), rendering zinc and magnesium comparatively brittle unless heated to activate non-basal pyramidal slip!

3. Slip in Single Crystals & Schmid's Law

Even when a tensile specimen is pulled under pure uniaxial normal stress ($\sigma = F/A_0$), dislocations within individual grains experience shear stresses along their inclined slip planes.



Mathematical Derivation of Schmid's Law

Consider a cylindrical single crystal with cross-sectional area A_0 loaded with tensile force F :

1. Let ϕ be the angle between the normal to the slip plane ($\mathbf{n}$) and the tensile loading axis.
2. Let λ be the angle between the slip direction ($\mathbf{s}$) and the tensile loading axis.
3. The component of the applied force projected parallel to the slip direction is:

$$F_s = F \cos \lambda$$

4. The area of the inclined slip plane is geometrically enlarged:

$$A_s = \frac{A_0}{\cos \phi}$$

5. The **Resolved Shear Stress** (τ_R) acting on the slip system is the projected force divided by the inclined slip plane area:

$$\tau_R = \frac{F_s}{A_s} = \frac{F \cos \lambda}{\frac{A_0}{\cos \phi}} = \left(\frac{F}{A_0} \right) \cos \phi \cos \lambda = \sigma \cdot m$$

Where:

- $m = \cos \phi \cos \lambda$ is defined as the **Schmid Factor**.
- Since the angle between the normal $\mathbf{n}$ and the direction $\mathbf{s}$ on the same plane is 90° , the sum of angles satisfies $\phi + \lambda \geq 90^\circ$.

Critical Resolved Shear Stress (τ_{crss})

Plastic deformation (yielding) initiates when the resolved shear stress on the most favorably oriented slip system reaches a material-specific threshold: the **Critical Resolved Shear Stress** (τ_{crss}):

$$\tau_R = \tau_{\text{crss}} \implies \sigma_y = \frac{\tau_{\text{crss}}}{\cos \phi \cos \lambda} = \frac{\tau_{\text{crss}}}{m}$$

Maximum Schmid Factor (m_{max})

To minimize the required external tensile yield stress σ_y , the Schmid factor must be maximized:

- Maximum value occurs when $\phi = \lambda = 45^\circ$:

$$m_{\text{max}} = \cos 45^\circ \cdot \cos 45^\circ = \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = \mathbf{0.50}$$

- Therefore, the absolute minimum applied tensile stress required to produce slip is:

$$\sigma_{y,\text{min}} = \frac{\tau_{\text{crss}}}{0.50} = 2, \tau_{\text{crss}}$$

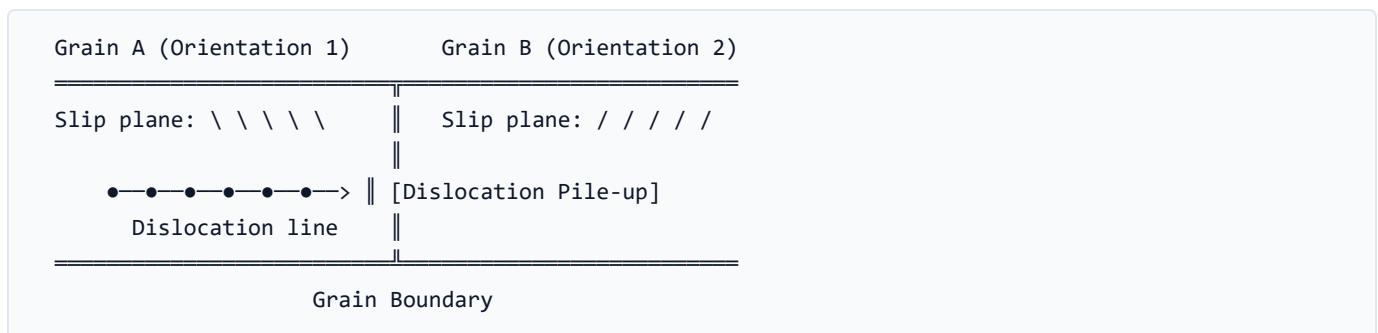
- If $\phi = 90^\circ$ (tensile axis in the slip plane) or $\lambda = 90^\circ$ (tensile axis perpendicular to the slip direction), $m = 0 \implies \tau_R = 0$, meaning no slip can occur on that system regardless of the applied load!

4. The Four Foundational Strengthening Mechanisms

To increase the yield strength σ_y and hardness of a metallic material, we must create barriers that impede dislocation motion. The four classic mechanisms are:

Mechanism 1: Grain Size Reduction (Hall-Petch Strengthening)

In a polycrystal, individual grains are separated by **grain boundaries**—narrow zones of atomic misalignment (2 – 3 atom diameters wide).



Grain boundaries serve as formidable barriers to slip for two reasons:

1. **Crystallographic Misorientation:** When a dislocation reaches a boundary, it cannot glide straight across into the neighboring grain because the slip planes and directions in Grain B do not align with those in Grain A. It must abruptly change direction, requiring substantial additional stress.
2. **Disordered Boundary Zone:** The atomic disorder at the grain boundary creates an absence of regular slip planes.

When dislocations are blocked, they pile up behind the boundary. The stress concentration at the head of the pile-up eventually triggers new dislocation sources in the adjacent grain. Finer grains produce shorter pile-ups, requiring a higher external stress to activate slip across the boundary!

The Hall-Petch Equation

$$\sigma_y = \sigma_0 + k_y \cdot d^{-1/2}$$

Where:

- σ_y = Yield strength of the polycrystal.
- σ_0 = Friction stress (yield strength of a single crystal / lattice friction).
- k_y = Hall-Petch strengthening coefficient (material constant measuring boundary resistance).
- d = Average grain diameter (m or mm).

[!TIP] **The Unique Superpower of Grain Refinement:** Grain size reduction is the **ONLY strengthening mechanism** in metallurgy that simultaneously **increases yield strength AND increases toughness/ductility!** All other mechanisms (cold working, alloying, precipitation) increase strength at the cost of reduced ductility.

Mechanism 2: Solid Solution Strengthening

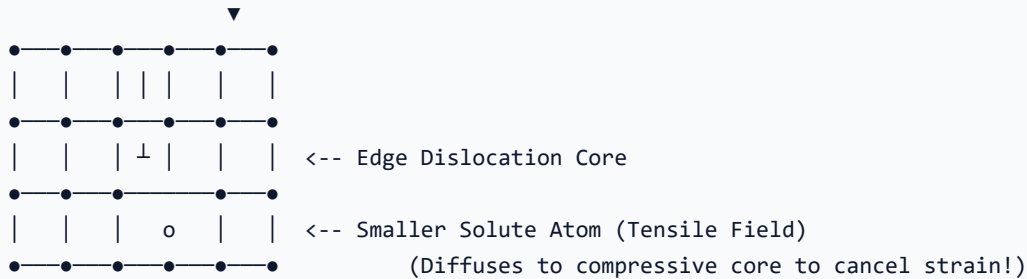
Alloying high-purity metals with solute atoms (e.g. adding Zn to Cu to make brass, or adding C to Fe to make steel) produces a substantial increase in strength and hardness.

Physical Mechanism: Lattice Misfit Strain Interaction

Because solute atoms differ in atomic radius from solvent atoms:

1. **Substitutional Solute Smaller than Solvent:** Surrounding host atoms collapse slightly inward, producing a localized **tensile strain field**.
2. **Substitutional Solute Larger than Solvent:** Host atoms are pushed outward, creating a localized **compressive strain field**.

Compressive Core Strain



- The compressive strain field of a large solute atom naturally migrates to the **tensile region** below an edge dislocation, canceling part of the lattice strain energy!
- Similarly, smaller solute atoms settle in the **compressive region** above the dislocation.
- This mutual strain relief locks (pins) the dislocation in a thermodynamic energy well. To move the dislocation, an increased external shear stress must be applied to tear it away from the solute atmosphere.

Mechanism 3: Strain Hardening (Cold Work)

Strain hardening (also called **work hardening** or **cold work**) is the phenomenon whereby a ductile metal becomes stronger and harder as it is plastically deformed at temperatures below its recrystallization temperature (typically at room temperature).

Quantitative Metric: Percent Cold Work (%)

Cold work is quantified by the percentage reduction in cross-sectional area:

Where:

- A_0 = Original cross-sectional area prior to deformation.
- A_d = Deformed cross-sectional area after cold rolling, drawing, or swaging.

Dislocation Multiplication Mechanism

Why does deforming a metal make it harder?

1. In an annealed (undeformed) metal crystal, the dislocation density is relatively low: $\rho_d \approx 10^5 - 10^6 \text{ cm/cm}^3$.
2. As the metal is rolled or forged, existing dislocations glide and interact with crystal defects, generating massive numbers of new dislocations via **Frank-Read sources**.
3. In a heavily cold-worked metal, dislocation density skyrockets by 4 to 5 orders of magnitude:

$$\rho_d \approx 10^9 - 10^{10} \text{ cm/cm}^3 \quad (\approx 10,000 \text{ km of dislocation lines per cm}^3!)$$

4. As dislocation density increases, the average spacing between dislocations shrinks drastically. Dislocation stress fields mutually intersect and entangle, forming dense, impenetrable tangles, sessile jogs, and cellular sub-boundaries.

5. Because dislocations mutually obstruct one another's motion, the shear stress required to produce further slip rises steadily according to the **Taylor equation**:

$$\tau_y = \tau_0 + \alpha \cdot G \cdot b \cdot \sqrt{\rho_d}$$

- Macroscopic consequence: **Yield strength σ_y increases, Tensile strength UTS increases, Hardness increases**, but **Ductility () decreases drastically** (the material becomes work-exhausted).

Mechanism 4: Precipitation Hardening (Dispersion Strengthening)

In aerospace alloys (e.g. 2000 and 7000 series aluminum), extremely high strength is achieved by heat treating (solutionizing, quenching, and artificial aging) to precipitate billions of nanoscale, hard second-phase particles (e.g. CuAl_2 or MgZn_2) uniformly throughout the matrix.

Dislocations encounter these precipitate particles as obstacles:

1. **Particle Cutting (Shearing)**: When precipitates are very small and coherent, the dislocation cuts directly through the particle, creating new interfacial surface energy.
2. **Orowan Bowing (Looping)**: When precipitates are larger and incoherent, dislocations cannot cut through them. The dislocation line bows between the particles until the loops pinch off, leaving a residual dislocation loop around each particle:

$$\tau_{\text{Orowan}} \approx \frac{G \cdot b}{L}$$

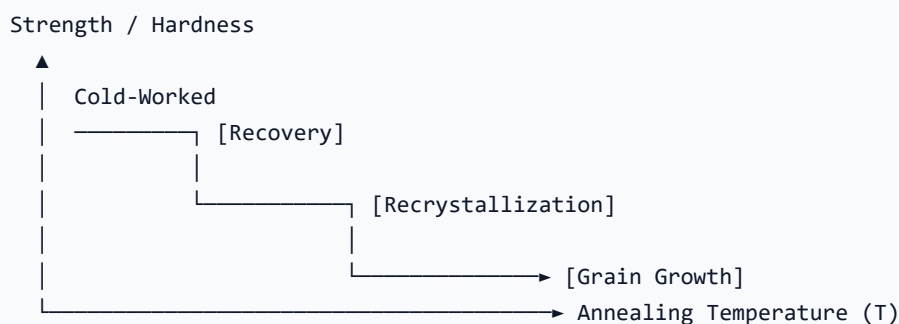
Where L is the interparticle spacing. Finer particle dispersions (smaller L) yield vastly higher strength!

5. Annealing of Cold-Worked Metals

When a metal is heavily cold-worked, approximately 95 of the mechanical work input is dissipated as heat, while the remaining 5 is stored internally as **elastic strain energy** associated with the massive tangle of dislocations.

This stored strain energy leaves the metal in a thermodynamically metastable, high-energy state. If the cold-worked metal is heated (annealed), atomic diffusion accelerates, allowing the material to revert to its low-energy, ductile state.

Annealing proceeds through three distinct, consecutive stages:



Stage 1: Recovery ($T \lesssim 0.3, T_m$)

- **Physical Process:** Enhanced thermal vibrations allow vacancies and atoms to diffuse over short distances. Dislocations rearrange themselves, annihilating opposite-sign pairs and polygonizing into organized low-energy, low-angle subgrain boundaries.
- **Property Changes:**
 - Residual internal stresses are significantly relieved.
 - Electrical and thermal conductivities are restored to nearly annealed levels.
 - **Microstructure and mechanical strength remain essentially unchanged** (no new grains form).

Stage 2: Recrystallization ($T_R \approx 0.3 - 0.4, T_m$ in Kelvin)

- **Physical Process:** Driven by the massive stored strain energy of cold work, a completely new set of **strain-free, equiaxed grains** nucleate and grow rapidly, consuming the deformed, high-dislocation cold-worked grains.
- **Property Changes:**
 - Dislocation density drops dramatically back to $\sim 10^5 - 10^6 \text{ cm}^{-2}$.
 - **Yield strength and tensile strength drop sharply.**
 - **Ductility is fully restored** to pre-cold-work levels!
- **The Recrystallization Temperature (T_R):** The temperature at which recrystallization reaches completion in exactly 1 hour.
 - For pure metals: $T_R \approx 0.3 - 0.4, T_m$.
 - Higher prior to annealing provides a greater driving force, lowering T_R and refining the recrystallized grain size!

Stage 3: Grain Growth ($T \gg T_R$)

- **Physical Process:** After recrystallization is complete, the strain-free grains continue to coarsen if held at elevated temperatures. Larger grains grow at the expense of smaller grains to minimize the total grain boundary surface area (reducing total interfacial energy).
- **Kinetic Law:** The average grain diameter d grows with annealing time t according to:

$$d^n - d_0^n = K \cdot t$$

Where $n \approx 2$ and K follows an Arrhenius temperature dependence ($K = K_0 \exp[-Q/RT]$).

6. Fully Solved Quantitative Archetypes

Problem 1: Schmid's Law & Yield Stress in a Single Crystal

Problem Statement: A single crystal of an FCC metal is loaded in uniaxial tension along its $[001]$ axis. Slip occurs on the (111) plane along the $[10\bar{1}]$ slip direction. The critical resolved shear stress is $\tau_{\text{crss}} = 0.80 \text{ MPa}$.

1. Calculate the angle ϕ between the tensile axis $[001]$ and the slip plane normal $[111]$.
2. Calculate the angle λ between the tensile axis $[001]$ and the slip direction $[10\bar{1}]$.
3. Determine the Schmid factor m .

4. Calculate the applied tensile stress σ_y required to initiate plastic deformation.

Step-by-Step Solution:

Step 1: Calculate $\cos \phi$ Using the vector dot product formula $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$:

- Tensile axis $\mathbf{u} = [001]$: $|\mathbf{u}| = \sqrt{0^2 + 0^2 + 1^2} = 1$.
- Slip plane normal $\mathbf{v} = [111]$: $|\mathbf{v}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$.

$$\cos \phi = \frac{(0)(1) + (0)(1) + (1)(1)}{(1)(\sqrt{3})} = \frac{1}{\sqrt{3}} \approx 0.5774 \implies \phi \approx 54.74^\circ$$

Step 2: Calculate $\cos \lambda$

- Tensile axis $\mathbf{u} = [001]$: $|\mathbf{u}| = 1$.
- Slip direction $\mathbf{w} = [10\bar{1}]$: $|\mathbf{w}| = \sqrt{1^2 + 0^2 + (-1)^2} = \sqrt{2}$.

$$\cos \lambda = \frac{(0)(1) + (0)(0) + (1)(-1)}{(1)(\sqrt{2})} = \frac{-1}{\sqrt{2}}$$

Taking the acute angle magnitude:

$$\cos \lambda = \frac{1}{\sqrt{2}} \approx 0.7071 \implies \lambda = 45^\circ$$

Step 3: Calculate the Schmid Factor (m)

$$m = \cos \phi \cdot \cos \lambda = \left(\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{\sqrt{6}} \approx 0.4082$$

Step 4: Compute Tensile Yield Stress (σ_y)

$$\sigma_y = \frac{\tau_{\text{CRSS}}}{m} = \frac{0.80 \text{ MPa}}{0.4082} = 1.96 \text{ MPa}$$

Problem 2: Hall-Petch Grain Size Scaling Calculation

Problem Statement: A brass alloy has an average grain diameter $d_1 = 100 \mu\text{m}$ (0.100 mm) and exhibits a yield strength $\sigma_{y1} = 120 \text{ MPa}$. When severely plastically deformed and recrystallized to $d_2 = 25 \mu\text{m}$ (0.025 mm), the yield strength increases to $\sigma_{y2} = 180 \text{ MPa}$.

- Determine the Hall-Petch constants σ_0 and k_y .
- Predict the yield strength if the grain size is further refined to $d_3 = 4 \mu\text{m}$.

Step-by-Step Solution:

Step 1: Convert grain diameters to $d^{-1/2}$ in $\text{mm}^{-1/2}$

- $d_1 = 0.100 \text{ mm} \implies d_1^{-1/2} = (0.100)^{-1/2} = \frac{1}{\sqrt{0.1}} \approx 3.1623 \text{ mm}^{-1/2}$

- $d_2 = 0.025 \text{ mm} \implies d_2^{-1/2} = (0.025)^{-1/2} = \frac{1}{\sqrt{0.025}} \approx 6.3246 \text{ mm}^{-1/2}$

Step 2: Set up system of linear equations

$$\sigma_{y1} = \sigma_0 + k_y(3.1623) = 120 \text{ MPa}$$

$$\sigma_{y2} = \sigma_0 + k_y(6.3246) = 180 \text{ MPa}$$

Subtracting the first equation from the second:

$$k_y(6.3246 - 3.1623) = 180 - 120$$

$$k_y(3.1623) = 60 \implies k_y = \frac{60}{3.1623} = \mathbf{18.97 \text{ MPa} \cdot \text{mm}^{1/2}}$$

Solving for σ_0 :

$$\sigma_0 = 120 - (18.97)(3.1623) = 120 - 60 = \mathbf{60.0 \text{ MPa}}$$

Step 3: Predict σ_y at $d_3 = 4 \text{ } \mu\text{m} = 0.004 \text{ mm}$

$$d_3^{-1/2} = (0.004)^{-1/2} = \frac{1}{\sqrt{0.004}} \approx 15.8114 \text{ mm}^{-1/2}$$

$$\sigma_{y3} = \sigma_0 + k_y \cdot d_3^{-1/2} = 60.0 + (18.97)(15.8114) = 60.0 + 300.0 = \mathbf{360.0 \text{ MPa}}$$

Problem 3: Cold Work Reduction in Rod Drawing

Problem Statement: A cylindrical copper rod originally having a diameter $d_0 = 15.0 \text{ mm}$ is cold drawn through a die to a final diameter $d_d = 12.0 \text{ mm}$. Calculate the percent cold work ().

Step-by-Step Solution:

7. Exam Pitfalls & High-Yield Summary Table

Concept	Governing Equation	Key Rule / Mechanism	Common Exam Pitfall
Schmid's Law	$\tau_R = \sigma \cos \phi \cos \lambda$	Slip occurs when $\tau_R \geq \tau_{\text{crss}}$. Max $m = 0.5$ at 45° .	Confusing slip plane angle with slip direction angle.
FCC Slip Systems	$111\langle 110 \rangle$ (12)	4 planes $\times$ 3 directions = 12 systems. High ductility.	Forgetting FCC has 12 systems, not 4.
BCC Slip Systems	$110\langle 111 \rangle$ (48)	48 systems, but lacks close-packed planes; prone to DBTT.	Believing 48 systems guarantees cryogenic ductility.
HCP Slip Systems	$0001\langle 11\bar{2}0 \rangle$ (3)	Only 3 basal systems at room temperature; brittle.	Failing the 5 independent systems Von Mises rule.
Hall-Petch	$\sigma_y = \sigma_0 + k_y d^{-1/2}$	Finer grains block dislocations. Increases strength AND ductility!	Forgetting the square root power (scaling with $d^{-1/2}$).
Solid Solution	Solute size misfit Δr	Solute strain fields pin dislocation cores.	Assuming chemical bonding changes; it is elastic strain!
Cold Work (%)		Dislocation multiplication from 10^6 to 10^{10} cm^{-2} .	Calculating linear diameter reduction instead of area!
Recovery	$T < T_R$	Stress relief, polygonization; NO new grains form.	Confusing recovery with recrystallization.
Recrystallization	$T_R \approx 0.3 - 0.4, T_m$	New strain-free equiaxed grains form; ductility restored.	Calling it phase change (crystal structure is identical).